

HR Analytics Project
Absenteeism Analysis & Compensation Strategy
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1. Problem Statement

The HR department has requested a data-driven analysis to support strategic decisions regarding employee wellness and absenteeism. The analysis must address the following key objectives:

- **Healthy Bonus Program:** Identify employees with healthy lifestyle habits and low absenteeism for a \$1,000 USD bonus pool.
 - **Non-Smoker Compensation Plan:** Allocate a \$983,221 USD insurance budget to reward non-smoking employees.
 - **Absenteeism Dashboard:** Create a dashboard to visualize absenteeism trends, reasons, and lifestyle factors.
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2. Project Overview

This project analyzes employee absenteeism data to uncover behavioral patterns, optimize compensation strategies, and support HR decision-making. By integrating multiple datasets and applying business logic, we identify key trends and recommend targeted interventions.

3. Data Sources

- absenteeism_at_work: Employee attendance records and lifestyle indicators.
 - compensation: Salary and financial data.
 - reasons: Maps numeric absence codes to descriptive reasons.
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4. Data Integration

To create a unified dataset, we performed SQL joins across the three tables:

```
SELECT *
```

```
FROM absenteeism_at_work a
```

```
LEFT JOIN compensation c ON a.ID = c.ID
```

```
LEFT JOIN reasons r ON a.`reason for absence` = r.number;
```

This enriched view allows for deeper analysis of absenteeism in relation to compensation and absence reasons.

5. Identifying Healthy Employees for Bonus

Criteria:

- Non-smokers
- Non-drinkers
- BMI < 25
- Absenteeism below average

```
SELECT *
```

```
FROM absenteeism_at_work
```

```
WHERE `Social drinker` = 0
```

```
AND `Social smoker` = 0
```

```
AND `Body mass index` < 25
```

```
AND `absenteeism time in hours` < (
```

```
SELECT AVG(` absenteeism time in hours ` ) FROM absenteeism_at_work  
);
```

6. Non-Smoker Compensation Plan

To estimate the budget required for rewarding non-smokers:

- Budget: \$983,221 USD
- Rate: \$0.68/hour or \$1,414.4/year

```
SELECT COUNT(*) AS non_smokers  
FROM absenteeism_at_work  
WHERE ` Social smoker ` = 0;
```

7. Optimized Analytical Query

This query categorizes BMI, identifies seasonal absence trends, and extracts relevant employee attributes:

```
SELECT  
a.ID,  
r.reason,  
` Month of absence ` ,  
` Body mass index ` ,  
CASE  
WHEN ` Body mass index ` < 18.5 THEN 'Underweight'  
WHEN ` Body mass index ` BETWEEN 18.5 AND 25 THEN 'Healthy'  
WHEN ` Body mass index ` BETWEEN 25 AND 30 THEN 'Overweight'  
WHEN ` Body mass index ` > 30 THEN 'Obese'  
ELSE 'Unknown'  
END AS Bmi_category,  
CASE
```

```

    WHEN `Month of absence` IN (12,1,2) THEN 'Winter'
    WHEN `Month of absence` IN (3,4,5) THEN 'Spring'
    WHEN `Month of absence` IN (6,7,8) THEN 'Summer'
    WHEN `Month of absence` IN (9,10,11) THEN 'Fall'
    ELSE 'Unknown'
END AS Season_names,
`Day of the week`,
`Transportation expense`,
Education,
son,
`Social drinker`,
`Social smoker`,
pet,
`Disciplinary failure`,
Age,
`Work load Average/day`,
`absenteeism time in hours`
FROM absenteeism_at_work a
LEFT JOIN compensation c ON a.ID = c.ID
LEFT JOIN reasons r ON a.`reason for absence` = r.number;

```

8. Dashboard Insights

From the dashboard analysis:

- Average absenteeism: 5.96 hours
- Top absentee: Employee 11 (42.46% of total hours)
- Seasonal trends: Absenteeism peaks in Fall and Winter
- Common reasons: Medical consultation, dental issues, musculoskeletal problems

- Behavioral factors: Smoking and drinking correlate with higher absenteeism

9. Recommendations

- Reward healthy behavior: Use bonuses to incentivize wellness.
- Target interventions: Focus on high-absence months and top reasons.
- Monitor lifestyle factors: Smoking and drinking are strong absenteeism indicators.

Final dashboard:

